Chapter 1: Network Layer and Routing Protocols

1.1 IP Addressing and Subnetting

The Internet Protocol (IP) address is a numerical label assigned to each device connected to a computer network. An IPv4 address is a 32-bit number that uniquely identifies a network interface on a host.

Subnetting divides a larger network into smaller, distinct logical network segments.

Chapter 2: Transport Layer and TCP/UDP

2.1 Transmission Control Protocol (TCP)

TCP is a connection-oriented, reliable transport layer protocol that provides in-order byte stream delivery. TCP employs a three-way handshake (SYN, SYN-ACK, ACK) to establish a connection before data transmission begins. Flow control is maintained using a sliding window mechanism.